SOUTH CITY™
INTERNATIONAL SCHOOL

FIRST TERMINAL MODEL PAPER
COMPUTER SCIENCE PAPER-1 (THEORY)
CLASS - XI
SESSION 2025-26

Time: 3 hours **Maximum Marks: 70**

(This Question Paper has 7 printed pages Candidates are allowed additional 15 minutes for only reading the paper.
They must NOT start writing during this time).

Answer all questions in Part I (compulsory) and six questions from Part-II, choosing two questions from Section-A, two from Section-B and two from Section-C.
All working, including rough work, should be done on the same sheet as the on the same sheet as the rest of the answer.
The intended marks for questions or parts of questions are given in brackets [ ]

PART I (20 Marks)

Answer all questions.

While answering questions in this Part, indicate briefly your working and reasoning, wherever required.

Question 1

i) When we convert 10010 binary numbers to decimals. Then the solution is : [1]

(a) 2010

(b) 1810

(c)1410

(d)1610

(ii) The NOR gate is the complement of : [1]

(a) NOT gate

(b) AND gate

(c) OR gate

(d) XOR gate

(iii) The complement of the Boolean expression $(P \cdot Q)' + R'$ is: [1]

(a) (P+Q).R

(b) PQR

(c) (P’+Q’) .R’

(d) (P’+Q’).R

(iv) The binary equivalent of the decimal number 0.4375 is : [1]

(a) 0.1010

(b) 0.1100

(c) 0.1011

(d) 0.0111

(v) Phenomenon of acquiring the base class properties and functions is known as: [1]

(a) Polymorphism

(b) Encapsulation

(c) Abstraction

(d) Inheritance

(vi) State Involution law and prove it with the help of a truth table. [2]

(vii) Show that $X \vee \sim (Y \wedge X)$ is a tautology. [1]

(viii) Find the dual of

YX + X’ + 1 = 1 [2]

Question 2

(i) The following function extractPart() is a part of some class which extracts a substring from a string text, starting from index start and extracting exactly count characters. Some parts of the code are missing and marked as ?1?, ?2?, ?3?.

Write the correct expressions/statements to complete the function.

```
String extractPart(String text, int start, int count)
{
   int i = ?1?;
   int limit = start + count;
   String result = "";

   while(?2? && i < text.length())
   {
      result += text.charAt(i);
      ?3?;
   }
   return result;
```

}

(a) What is the expression or statement at ?1? [1]
(b) What is the expression or statement at ?2? [1]
(c) What is the expression or statement at ?3? [1]

ii)a) Given that int x[ ][] = ({2,4,6), 13, 5, 7}; what will be the value of x[1][0] and x[0][2]? [1]

b) What is the value of x[0].length [1]

(iii) Write the Java statement to initialize character array. [1]

iv) Give the output of the following statements [3]

a) System.out.print("abe". indexOf('b'));
(b) System.out.print("S".tolowerCase);
(c) System.out.print("India".compareTo("country");

v) Consider the following code and answer the questions that follow

```
int x, y;
void access( )
int a, b;
academic student = new academic( );
System.out.println ("object created");
```

a)What is the object name of class academic ? [1]
b) Name the class variables used in the program. [1]

PART – II(50 Marks)

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C.

SECTION - A

Answer any two questions

Question 3

(i) Prove the following using algebraic method.

$[(p \Rightarrow q) \wedge . \sim q] \Rightarrow \sim p$ [2]

(ii) Convert (AC26)16 to decimal. [1]

(iii) Convert (445.271)8 to binary. [1]

(iv) Convert (4896)10 to Hexadecimal. [1]

(v) Prove X + YZ = (X+Y)(X+Z) [2]

(vi) Construct the inverse, converse and contrapositive of : [3]

If it is January, then it is cold

Question 4

(a) Find the compliment of the following using De Morgan's Law. [2]

(A+B) . (A+C).B'

(b) Simplify the following expression using laws of Boolean algebra [2]

(A+B)`+ (A+B)'

(c) Using truth table prove that the following bi-conditional connective is a tautology. [2]

p=> q ⇔ p'V q

(d) Convert the following arithmetic expression into Java statement

x = (ax5 + bx7)/ √ab. [2]

(e) Differentiate between static variable and non static variable. [2]

Question 5

(a) Convert (22.25)10 to binary. [2]

(b) Subtract 10011 from 111101 [2]

(c) Obtain the truth table to verify the following expression: X(Y+Z) = XY + XZ. Also name the law stated above. [2]

(d) Answer the following questions related to the gate given below:

(e) i) What is the output of the above gate if input A=0, B=1 [1]

(f) ii) What are the values of the inputs if output is 1? [1]

(g) iii) Given F= A +(B=C) .(D'+E) . Find F' and show the relevant working in steps. [2]

SECTION – B
Answer any two questions.
Each program should be written in such a way that it clearly depicts the logic of the problem.
This can be achieved by using mnemonic names and comments in the program.
(Flowcharts and Algorithms are not required.)The programs must be written in Java.

Question 6

Define a class **Elect_Bill** having the following description:

- **Class name:** Elect_Bill

Data members / Instance variables:

int prev : to store the previous meter reading
int pres : to store the present meter reading
int units : to store the unit consumed (i.e. pres - prev)
String name : to store name of the consumer
double bill : to store the bill amount
double total : to store the total amount to be paid
Member functions / Methods:

void input() : to accept the name of the consumer along with previous and present reading

void cal() : to calculate the bill amount and total amount (along with monthly rent) to be paid

void display() : to display the name of the consumer, unit consumed, bill amount and total amount to be paid

The program will compute the monthly bill according to the following conditions and display the output as per the given format

Unit Consumed	**Rate**
Up to 100 units	No charge
Next 100 units	90 paise per unit
Next 200 units	80 paise per unit
More than 400 units	70 paise per unit

Additionally, every consumer has to pay ₹180 per month as monthly rent for the service.

Task:

- Specify the class Elect_Bill giving details of data members, methods, and logic for calculation.
- Define the main() function to create an object and display output in the following format:

Name of the customer   Unit Consumed   Total amount to be paid

------------------------   --------------   ------------------------

------------------------   --------------   ------------------------   [10]

Question 7

A class **DataStats** contains a single-dimensional list to store integers and perform various operations. Some of the members of the class are given below:

- **Class name:** DataStats

Data members / instance variables:

- nums[] : list to store integer elements
- n : to store the size of the list

Member functions / methods:

- DataStats(int nn) : parameterized constructor to initialize n = nn and declare the list
- void readData() : to accept the list elements from the user
- void findLargest() : finds and displays the largest element in the list with an appropriate message
- void displayData() : displays the list elements

Specify the class **DataStats**, giving the details of the **constructor**, void readData(), void findLargest(), and void displayData().
Also, define the **main()** function to create an object and call the functions accordingly to enable the task. [10]

Question 8

Write a program in Java to store 10 different numbers in a single-dimensional array and arrange them in such a way that all the even numbers are arranged in order first, followed by all the odd numbers.

Sample Input

25, 61, 15, 25, 59, 91, 82, 46, 64

Sample output

46, 64, 82, 15, 25, 25, 35, 59, 61, 91 [10]

SECTION – C
Answer any two questions.
Each program should be written in such a way that it clearly depicts the logic of the problem stepwise. This can be achieved by using comments in the program and mnemonic names or pseudo codes for algorithms.
The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.
(Flowcharts are not required).

Question 9

(a) State whether (p v q) => (p & q) is a tautology? [2]

(b) x= It is raining. y= It is not a sunny day. [3]

Construct the compound sentences for the following expressions:

i) ~ x => y

ii) x ⇔ y

iii) x+y

Question 10

(c) Define 'Base of number system? Give one example. [2]

(d) Draw a logic diagram for the following: ab + a\`b\` [3]

Question 11

A college is selecting candidates for a senior teaching position. The selection rules are:

- A candidate is selected if they are a Postgraduate (P) and have more than 15 years of teaching experience (E).

OR

- A candidate is an employee of the same college (S) and possesses a B.Ed. degree (B).

Inputs:

Inputs	
P	Candidate is a Postgraduate
S	Candidate is an employee of the same college
E	Candidate has teaching experience > 15 years
B	Candidate possesses a B.Ed. degree

Output:

C — Candidate is selected (1 = Selected, 0 = Not Selected)

Draw the truth table for inputs and outputs.
Derive the terms with conjunctive operators for each of the true value (1s) from the output column.
Also write the preposition expression by joining the terms with disjunction operators
[5]

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